

End-to-End Inventory Prediction and Contract Allocation for Guaranteed Delivery Advertising

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Background

Guaranteed Delivery (GD) Advertising is a strategic approach where advertisers secure their desired inventory of advertising impressions in advance by signing contracts with publishers weeks or months ahead of the targeting dates



GD Advertising places a fix price on impressions adding stability
It had better controls on allocation and targeting audience

Real-time Bidding (RTB) is dynamic and competitive, which advertisers don't like as much
RTB can be more computationally expensive

Allocation decision comes down to a bipartite graph problem where supply of users has to be matched to the demand from the advertisers for the impressions. This results in an elegant mathematical problem of prediction and optimization.

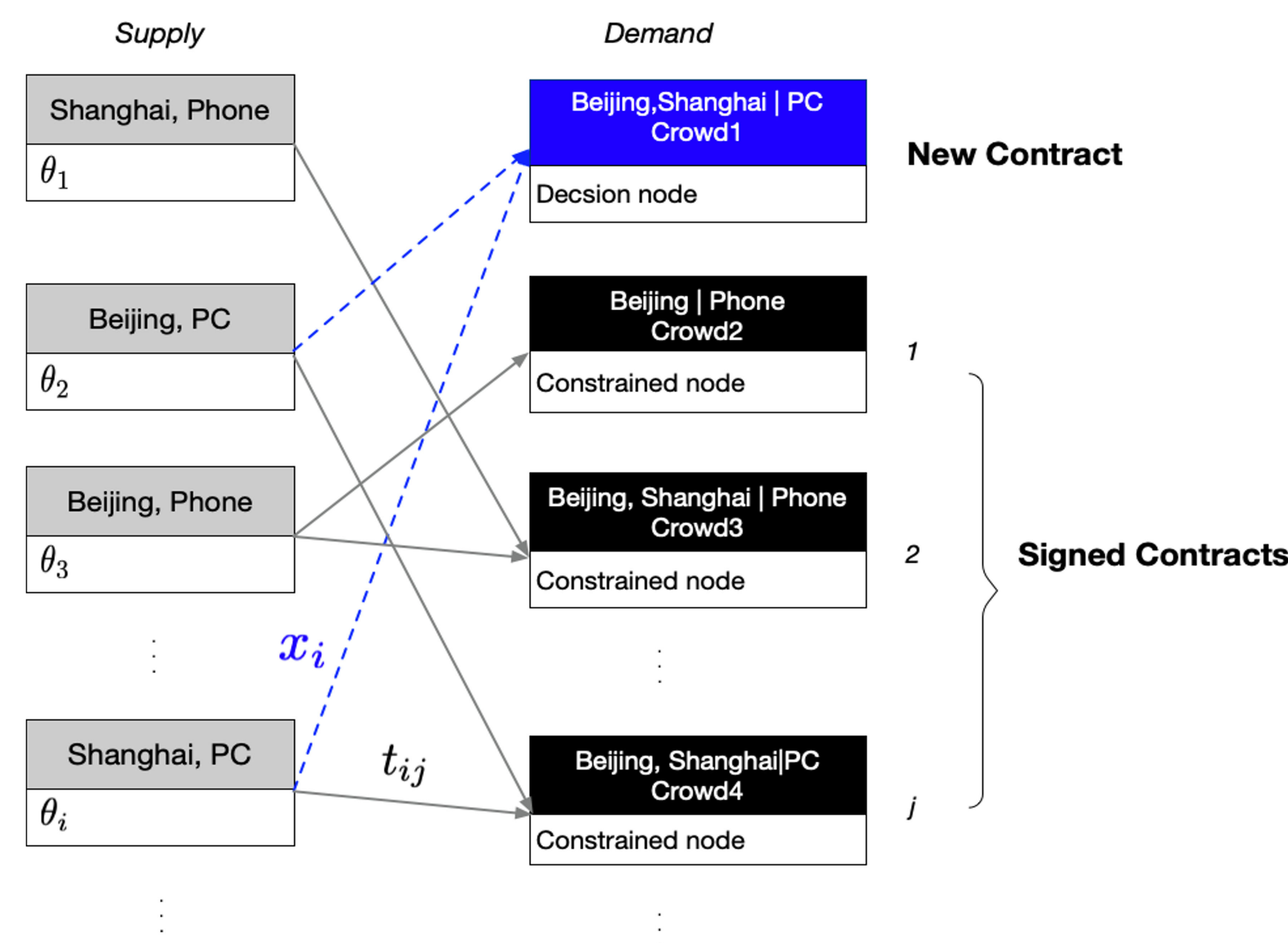
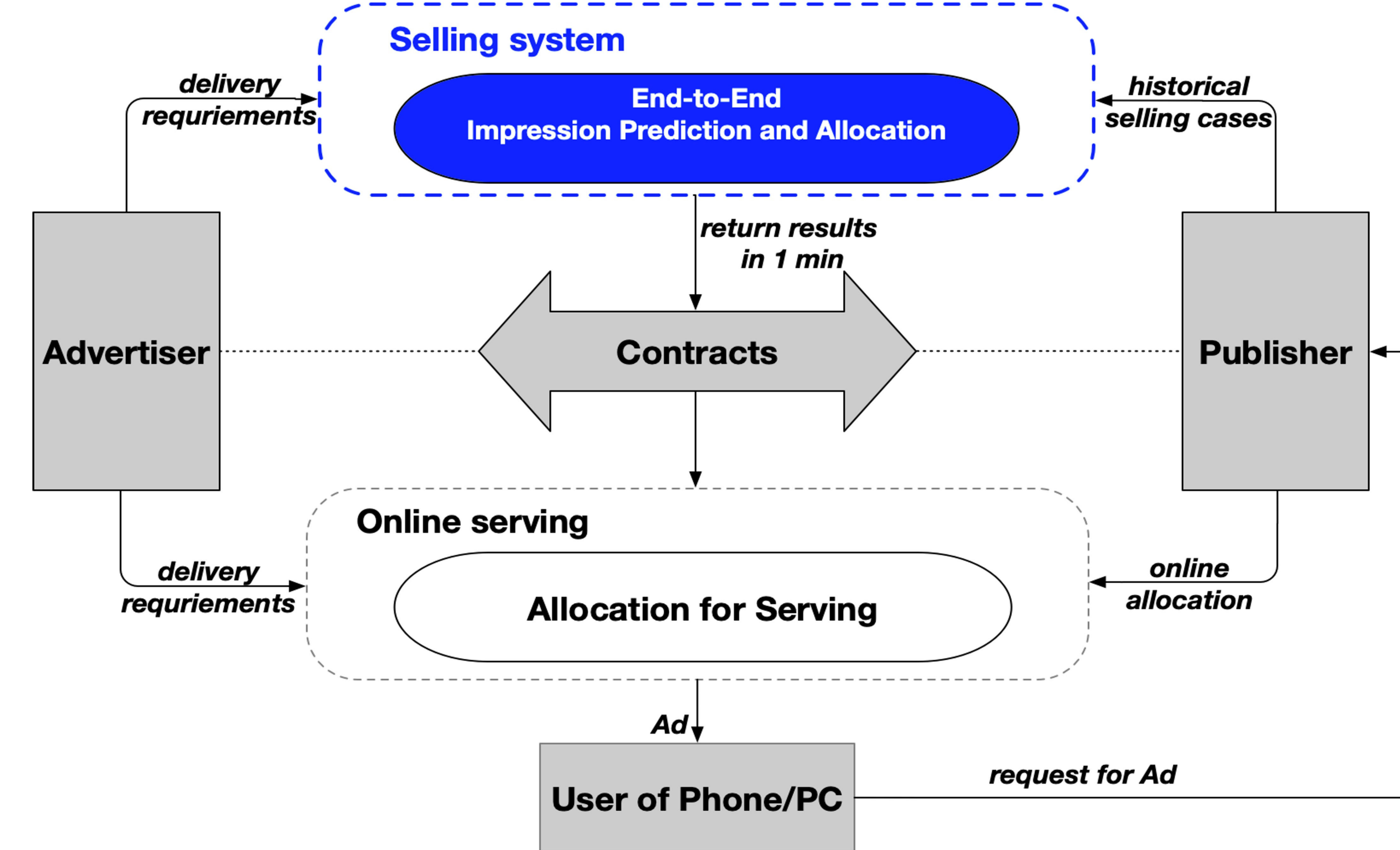


Figure: Bipartite graph of contract allocation problem. Supply nodes are impression inventories while demand nodes are impression contracts

System Architecture

System Architecture for Guaranteed Delivery (GD) Advertisement at Alibaba's websites where advertisers and publishers enter into a contract for impressions and have delivery requirements

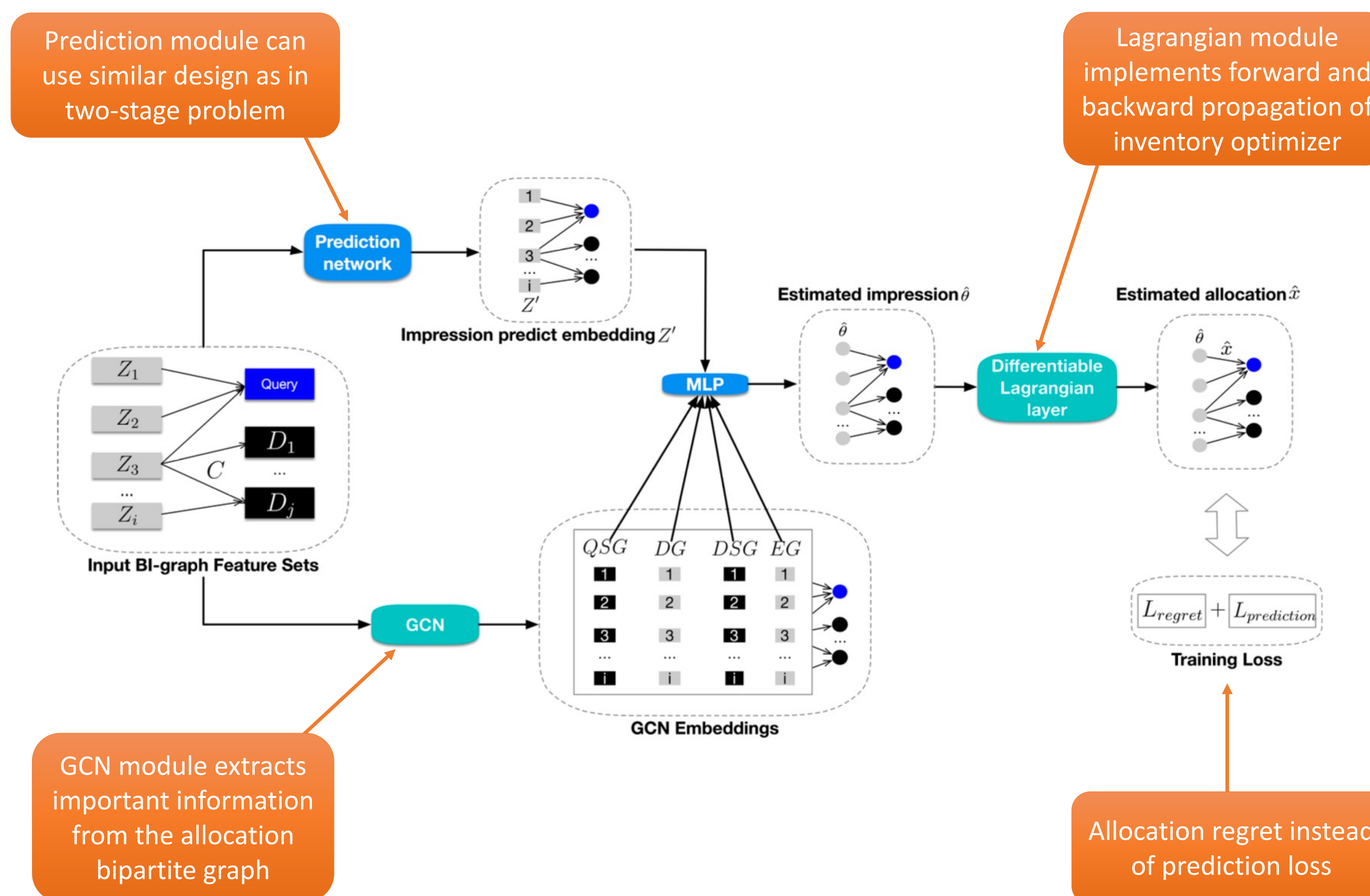


Neural Lagrangian Selling (NLS)

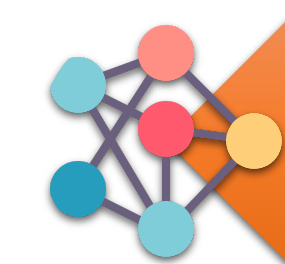
Traditionally, the problem has been solved in two stages:

1. Predicting the impressions from each user base using some predictive model
2. Optimizing allocation based on the predicted impressions using linear programming

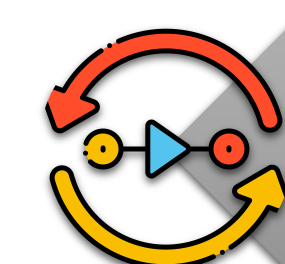
Our proposed solution (**Neural Lagrangian Selling**) blends the two stages into a single End-to-End Prediction and Optimization solution by minimizing allocation regret



Key Innovations



Graph Convolutional Networks (GCN) create embeddings responsible for constraints in our models



Differentiable Lagrangian Layer backpropagates allocation error to the neural network creating an end-to-end model

Results

Benchmark Models

Two-stage Model
• Compares end-to-end approach with traditional two-stage method
Pure Fully-Connected (PF)
• Baseline end-to-end approach using a simple black-box neural network
Pure Prediction GCN (PPG)
• Removes Lagrangian layer from NLS, uses GCN for prediction
Prediction Network + Lagrangian solver (PL)
• End-to-end approach without GCN module
GCN+QPTL/GCN+InOpt (End-to-End)
• LP solvers InOpt and QPTL to compare with Lagrangian layer

Evaluation Metrics

	End-to-End Normalized Deviation Error
	First-stage Error
	Second-stage Error
	Publisher Revenue (Avg. Revenue/Day)
	Delivery Rate (Delivered/Promised)
	Usage Rate (Sold/Available Impressions)

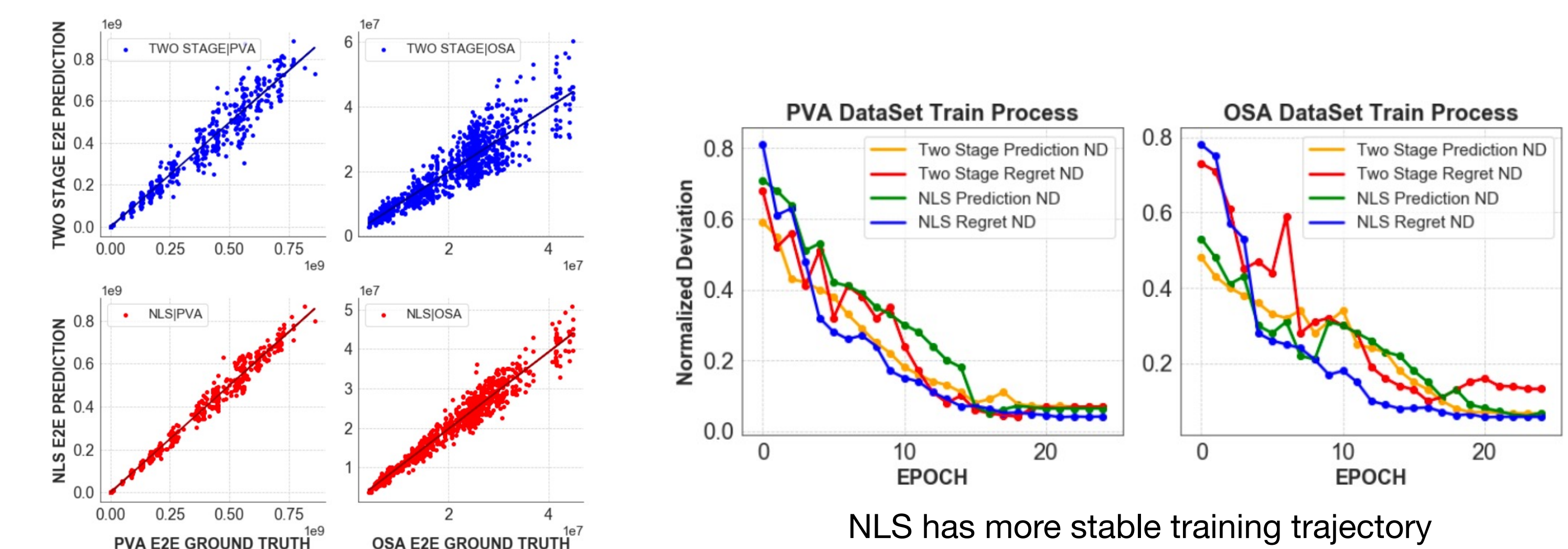
NLS outperforms other models on most benchmarks

Methods	full targeting		single targeting		random targeting	
	ND _{pre}	ND _{reg}	ND _{pre}	ND _{reg}	ND _{pre}	ND _{reg}
Two Stage	0.101±0.003	0.023±2e-4	0.101±0.003	0.125±0.005	0.101±0.003	0.045±0.002
PF	0.130±0.010	0.045±0.005	0.115±0.008	0.132±0.010	0.121±0.010	0.076±0.007
PPG	0.125±0.010	0.015±1e-4	0.127±0.007	0.112±0.001	0.135±0.011	0.036±0.001
PL	0.102±0.001	0.008±1e-4	0.103±0.001	0.113±0.003	0.101±0.002	0.047±2e-4
NLS	0.096±0.002	0.007±2e-4	0.097±0.001	0.098±0.001	0.095±0.001	0.029±1e-4

Experiment results on Offline data

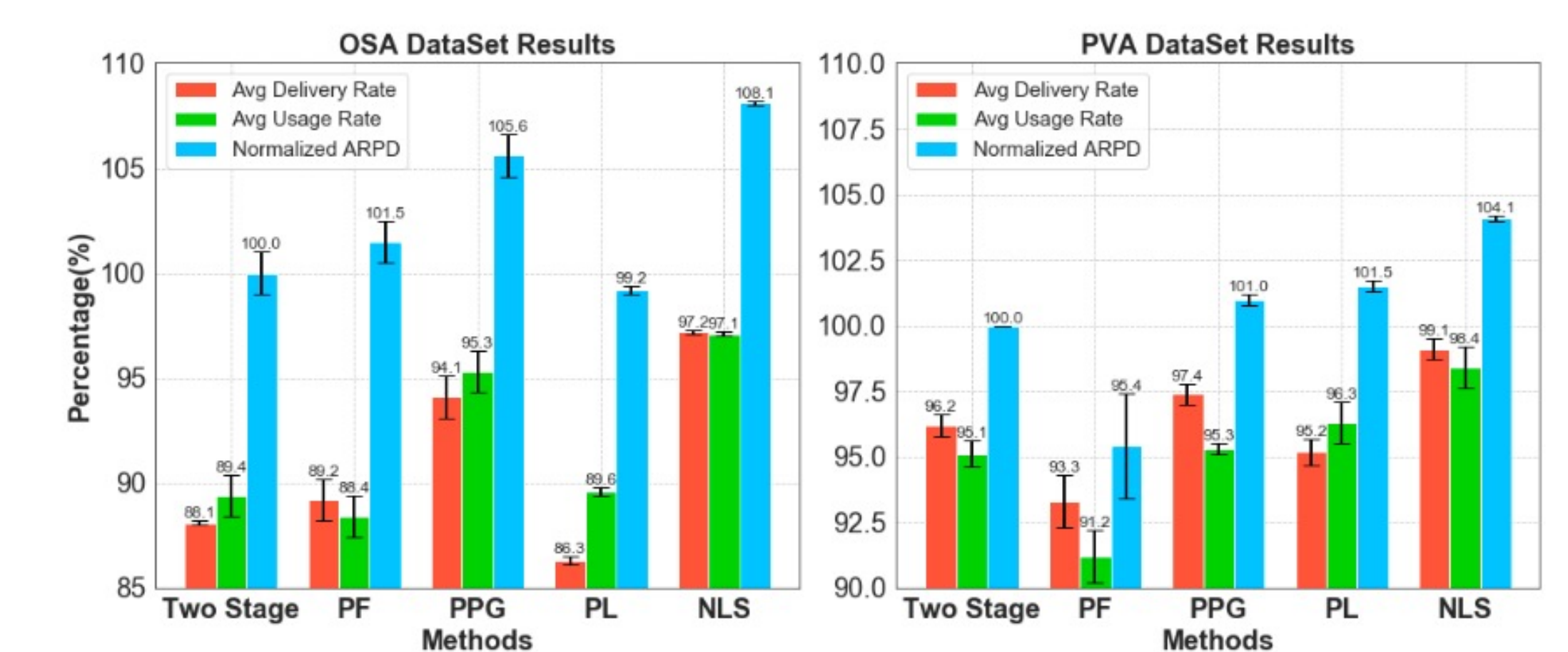
Methods	PVA		OSA	
	ND _{pre}	ND _{reg}	ND _{pre}	ND _{reg}
Two Stage	0.069±0.002	0.068±0.004	0.067±0.002	0.132±0.005
PF	0.083±0.015	0.095±0.020	0.075±0.015	0.128±0.021
PPG	0.085±0.005	0.054±0.002	0.078±0.003	0.086±0.004
PL	0.065±0.002	0.061±0.002	0.065±0.001	0.136±0.004
NLS	0.064±0.003	0.041±0.001	0.068±0.003	0.058±0.001

Experiment results on Online data



NLS has more stable training trajectory

NLS has fewer outliers in comparison with two-stage methods



NLS has better delivery rate, usage rate and publisher revenue